

HM2 Serie 2

$$1. a) f(x_1, x_2) = \begin{pmatrix} f_1(x_1, x_2) \\ f_2(x_1, x_2) \end{pmatrix} \quad \begin{aligned} f_1 &= 5x_1x_2 \\ f_2 &= x_1^2x_2^2 + x_1 + 2x_2 \end{aligned}$$

$$\begin{array}{ll} \text{PA: } f_1 & f_2 \\ x_1 \frac{\partial f_1}{\partial x_1} = 5x_2 & x_1 \frac{\partial f_2}{\partial x_1} = 2x_1x_2^2 + 1 \\ x_2 \frac{\partial f_1}{\partial x_2} = 5x_1 & x_2 \frac{\partial f_2}{\partial x_2} = 2x_1^2x_2 + 2 \end{array}$$

$$Df(x_1, x_2) = \begin{pmatrix} 5x_2 & 5x_1 \\ 2x_1x_2^2 + 1 & 2x_1^2x_2 + 2 \end{pmatrix} \quad \text{mit } \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$Df(1, 2) = \begin{pmatrix} 10 & 5 \\ 9 & 6 \end{pmatrix}$$

$$b) f(x_1, x_2, x_3) = \begin{pmatrix} f_1(x_1, x_2, x_3) \\ f_2(x_1, x_2, x_3) \\ f_3(x_1, x_2, x_3) \end{pmatrix}$$

PA: f_1

$$x_1 = \frac{1}{x_1^2 + x_2^2} \cdot 2x_1 + 0 = \frac{2x_1}{x_1^2 + x_2^2}$$

$$f_1 = \ln(x_1^2 + x_2^2) + x_3^2$$

$$f_2 = \exp(x_2^2 + x_3^2) + x_1^2$$

$$f_3 = \frac{1}{x_3^2 + x_1^2} + x_2^2$$

$$x_2 = \frac{1}{x_1^2 + x_2^2} \cdot 2x_2 + 0 = \frac{2x_2}{x_1^2 + x_2^2}$$

$$x_3 = 2x_3$$

PA: f_2

$$x_1 = 2x_1$$

$$x_2 = \exp(x_2^2 + x_3^2) \cdot 2x_2$$

$$x_3 = \exp(x_2^2 + x_3^2) \cdot 2x_3$$

PA: f_3

$$x_1 = -1(x_3^2 \cdot x_1^2)^{-2} \cdot 2x_1 = -\frac{2x_1}{x_3^2 \cdot x_1^2}$$

$$x_2 = 2x_3$$

$$x_3 = -1(x_3^2 \cdot x_1^2)^{-2} \cdot 2x_3 = -\frac{2x_3}{x_3^2 \cdot x_1^2}$$

$$Df(x_1, x_2, x_3) = \begin{pmatrix} \frac{2x_1}{x_1^2 + x_2^2} & \frac{2x_2}{x_1^2 + x_2^2} & 2x_3 \\ 2x_1 & \exp(x_2^2 + x_3^2) \cdot 2x_2 & \exp(x_2^2 + x_3^2) \cdot 2x_3 \\ -\frac{2x_1}{x_3^2 \cdot x_1^2} & 2x_3 & -\frac{2x_3}{x_3^2 \cdot x_1^2} \end{pmatrix}$$

$$\text{mit } \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

$$Df(1, 2, 3) = \begin{pmatrix} \frac{2}{5} & \frac{4}{5} & 6 \\ 2 & 4\exp(13) & 6\exp(13) \\ -\frac{1}{50} & 4 & -\frac{3}{50} \end{pmatrix}$$